

Regularising a CCD to full support

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A conditional clade distribution (CCD) assigns probability zero to any tree containing a clade or split its sample never produced. For held-out evaluation and for use as a tree prior this is too brittle: an independent posterior sample routinely visits trees just outside the support. We want a regulariser that gives *every* tree nonzero probability while keeping the observed structure sharp, and that places the right amount of mass on the unseen region. Here we develop a full-support, single-parameter regulariser that computes the probability of novelty at lazily without enlarging the clade set. Appendix A records an earlier route, NNI clade expansion, which enlarges the clade set by a fixed local move-set; the main model subsumes its coverage with a simpler construction.

CCD regularisation: full support from a single parameter

Motivation. One way to give unsampled clades nonzero probability is to *enlarge the clade set* with the clades reachable by a fixed local move (NNI clade expansion; Appendix A). Such an approach has three problems: it *mutates the graph* (a clade-enumeration pass plus a second split expansion), it adds extra parameters (β , mode) on top of α , and it covers only the novelty its move-set can reach so that trees more than one NNI from the support still receive probability zero. The construction we prefer instead computes a probability of unsampled clades on demand without touching the sampled clade set, gives *every* tree nonzero probability, and is governed by a single parameter μ , the per-clade probability of escaping the sampled splits.

Red/blue decomposition. Keep the split-expanded, additive- α backbone of (6), but add *no* clades. To score a tree T , colour each internal node *red* if its split is present in the (expanded) CCD and *blue* otherwise; a blue node necessarily involves an unobserved clade. Blue nodes form maximal connected *regions*. The top of a region is always an observed clade C (its parent is red, or it is the root), and its boundary ∂B is the set of m observed subclades hanging immediately below it (the first red or leaf nodes). Such a region resolves C into its m boundary parts through exactly $m - 2$ novel intermediate clades.

Probability. Red nodes keep their smoothed conditional clade probability θ , discounted by a reserved escape mass μ ; each blue region is charged one factor of ϵ per novel clade, spread uniformly over the ways its boundary can be realised:

$$P(T) = \prod_{v \in \text{red}(T)} (1 - \mu) \theta(S_v \mid C(v)) \prod_{B \in \text{blue}(T)} \frac{\epsilon(C_B)^{m_B - 2}}{\pi(C_B, \partial B)}, \quad (1)$$

where $\pi(C, \partial B)$ (the *pathcount*) is the number of all-novel resolutions of C into the boundary ∂B , computed by a subset dynamic program over the at most m parts. The division by π makes the reserved mass uniform over the unseen trees of a given boundary.

Per-clade reserve. The escape factor $\epsilon(C)$ is fixed per clade by requiring the blue mass to equal a target escape probability μ :

$$R(C) = \sum_{j \geq 1} N_j(C) \epsilon(C)^j = \mu, \quad N_j(C) = \#\{\text{partitions of } C \text{ into } j+2 \text{ observed subclades}\}, \quad (2)$$

restricted to partitions admitting an all-novel resolution (the pathcount cancellation in (1) collapses the resolutions, leaving one term per partition). Thus μ is the per-clade probability of escaping the sampled splits, and is the model's only free parameter; $\mu = 0$ recovers the unsmoothed backbone.

Full support by decoupling coverage from normalisation. Evaluating (2) exactly is #P-hard: $N_j(C)$ is a set-partition count over the non-laminar clade family, growing as $O(m^{j+1})$ in the number m of observed subclades of C . The key observation is that *coverage does not need it*. A tree receives nonzero probability as soon as (i) scoring is not truncated, and (ii) $\epsilon(C) > 0$ at each region top. Any region actually present in T is itself a valid novel partition, so $N_m(C) \geq 1$ and a positive ϵ exists. The exact N_j only fix the *value* of ϵ . We therefore solve (2) from the cheap low orders alone (reserve depth k ; N_1 is $O(m^2)$, N_2 is $O(m^3)$) and score regions of *every* depth with that ϵ . The result is full support based on a low-order approximation of $\epsilon(C)$ per sampled clade: every tree, at any novelty depth, is covered.

Normalisation and a non-inflating guarantee. Scoring all orders based on a depth- k estimate of $\epsilon(C)$ will assign slightly too much probability mass to trees that escape the CCD at clade C by omitting the tail, so the model is slightly *super-normalised*. This *inflates* the held-out log-probabilities and overestimates the fit of the model in any held-out comparison. The defect is

$$\text{tail}(C) = \sum_{j>k} N_j \epsilon^j \leq N_2 \epsilon^2 \frac{\rho}{1-\rho}, \quad \rho = \frac{N_2}{N_1} \epsilon, \quad (3)$$

a geometric upper bound read straight off the two computed orders (rigorous when the N_j are log-concave), scaling as μ^{k+1}/m^k . Discounting red splits by $1 - \mu - \text{tail}(C)$ instead of $1 - \mu$ guarantees the total probability can be at most 1: the model is then *sub-normalised* and can never over-state the model fit. A Knuth backtrack-tree estimate of the tail gives the near-exact value instead of the bound. On RSV2 the inflation scales as μ^3 and is negligible for realistic posterior samples, becoming material only for large μ :

μ	true inflation (sampled)	upper bound
0.01	0.0001	0.0001
0.05	0.0007	0.0101
0.10	0.014	0.077
0.20	0.116	0.662

(mean nat/tree on held-out RSV2). A safety check warns whenever $\mu > 0.05$, beyond which the bound exceeds ~ 0.01 nat/tree.

RSV2. Under 5-fold leave-one-tree-out on the strict-clock posterior ($n = 129$, 2,251 trees, backbone $\alpha = 0.4$, $\mu = 0.01$), the blue-region model covers every held-out tree at mean held-out log-probability competitive with the tuned graph-wide NNI of Appendix A:

model	coverage	mean log P (common set)
regCCD (no clade expansion)	89.6%	-53.50
NNI clade expansion (co-occurring)	99.3%	—
KRegCCD, ($\mu = 0.01$)	100.0%	-54.20

Full support guarantees the 100% structurally rather than by enlarging a clade set; the residual that NNI co-occurring misses (novelty an SPR-move away) is covered here automatically. The common-set log-probabilities are over the trees all models cover (the $\sim 90\%$ within plain regCCD’s support), on which regCCD is marginally sharper precisely because it reserves no escape mass there; the blue-region model trades that fraction of a nat to cover the remaining $\sim 10\%$ of held-out trees that contain a novel clade, which regCCD assigns probability zero.

Calibration: the right amount of novelty. Because μ is the only parameter and directly governs the escape mass, the model’s total probability on trees that contain a novel clade has a closed form,

$$P(\text{novel}) = 1 - M(\rho), \quad M(c) = (1 - \mu)^{[c \text{ reserves}]} \sum_{(L,R) \in S^{\text{obs}}(c)} \theta(L, R) M(L) M(R), \quad (4)$$

the mass that stays entirely within the observed clade set ($M = 1$ on leaves; the $(1 - \mu)$ factor applies at clades large enough to host an escape). Training on one half of the strict-clock posterior and scoring the independent held-out half, we compare (4) against the *empirical* fraction of held-out trees that actually contain a novel clade (0.214 here because the modest training sample makes this a high-novelty regime). Mean log-probability is over *all* held-out trees (the model covers them all), so it sits below the common-set value of the table above.

μ	model $P(\text{novel})$	empirical	ratio	mean log P/tree
0.002	0.132	0.214	0.62×	−56.953
0.003	0.192	0.214	0.90×	−56.869
0.0035	0.220	0.214	1.03×	−56.846
0.004	0.247	0.214	1.15×	−56.830
0.005	0.299	0.214	1.40×	− 56.816
0.006	0.347	0.214	1.62×	−56.818
0.010	0.509	0.214	2.38×	−56.907

The mean held-out log-probability has an interior optimum (too small a μ starves the genuinely novel trees, too large dilutes the observed ones), and is *flat* across $\mu \in [0.004, 0.006]$, all within ~ 0.005 nat of the peak at $\mu = 0.005$. The operating point can therefore be chosen for calibration at essentially no cost in sharpness: at $\mu \approx 0.004$ the model places mass on novel-clade trees within $\sim 1.15\times$ of the empirical rate (exact calibration, $1.03\times$, at $\mu \approx 0.0035$), giving up only ~ 0.01 nat against the sharpness peak. A single parameter, pinned inside a flat held-out-likelihood plateau, thus delivers both full coverage and the right amount of mass on the unseen region. This is the blue-region counterpart of the factor-1.2 calibration that Appendix A obtains from the two-parameter (α, β) smoothing, here from one parameter and the closed-form recursion (4).

Two weightings of the unseen region. The pathcount normaliser $\pi(C, \partial B)$ in (1), together with its matched partition count N_j in (2), encodes a modelling *choice* about how a region’s escape mass is divided among the distinct novel trees that realise it. Two self-consistent (properly normalised) weightings sit at the extremes:

- **Shared** (the form above). A boundary ∂B reserves a total $\epsilon(C)^{m-2}$, shared uniformly over its $\pi(C, \partial B)$ all-novel resolutions, so each tree is priced ϵ^{m-2}/π and the reserve counts *boundaries*, $N_j(C)$. Because π grows super-exponentially with the number of novel clades packed into one region ($\pi \sim (2k-3)!!$ for a region of k parts), this *penalises clustered novelty*: the marginal cost of adding a novel clade inside an existing region grows with the cluster.
- **Flat** (locally Boltzmann). Drop the π normaliser and price each distinct novel resolution at $\epsilon(C)^{m-2}$ directly; the reserve then counts novel *trees*,

$$R(C) = \sum_{j \geq 1} M_j(C) \epsilon(C)^j = \mu, \quad M_j(C) = \sum_{\partial B: |\partial B|=j+2} \pi(C, \partial B), \quad (5)$$

the number of all-novel resolutions of C with j novel intermediate clades. Each novel clade now costs a constant factor $\epsilon(C)$ irrespective of how many others share its region, so within a region $\log P$ falls by a constant $\log \epsilon$ per discordant clade: the prior is *locally Boltzmann in Robinson–Foulds distance* ($\epsilon(C)$ remains depth-dependent across regions, so it is not globally so).

Both weightings reserve the same total mass μ at each clade; they differ only in how it is spread. They coincide exactly on trees with no novel clade (scored on the red backbone alone) and at the single-novel-clade order ($M_1 \approx 2N_1$, so ϵ halves and the dropped $1/\pi$ cancels), and diverge only for clustered, multi-clade regions — precisely the trees a poorly sampled support produces.

Which weighting, and when. The choice is decided by how clustered the held-out novelty is, which is in turn set by how well the training posterior is sampled. Training a backbone ($\alpha = 0.4$, $k = 2$) on a leading block of the RSV2 posterior and scoring a disjoint held-out block, we bucket each held-out tree by its number of novel clades and report the flat–shared difference in mean held-out log-probability (nat/tree, at the selected μ):

training set	frac. ≥ 2 -novel	Δ_0	Δ_1	$\Delta_{\geq 2}$	Δ_{all}
saturated (7,200 trees)	0.5%	0.000	-0.25	+0.31	-0.005
sparse (1,000 trees)	6.0%	0.000	-0.18	+0.70	+0.018

Here Δ_b is the flat-shared difference restricted to held-out trees with b novel clades, and Δ_{all} is the overall mean. The difference lives entirely in the novel-clade trees: identically zero on the (majority) observed-clade trees, negative on single-novel-clade trees — where the flat weighting diverts mass toward clustered configurations — and strongly positive on clustered (≥ 2) trees, where it declines to penalise clustering. A well-sampled support produces almost no clustered novelty (0.5% of held-out trees), so shared is marginally sharper overall; a sparsely sampled support produces it routinely (6.0%), and flat wins by 0.018 nat/tree. Since regularisation is needed *precisely* when the support is incomplete, we take the flat (locally Boltzmann) weighting as the default.

Why this is preferred. The blue-region construction matches the NNI expansion’s coverage and sharpness while being strictly simpler and more general:

- **One parameter.** A single escape probability μ replaces $(\alpha_{\text{NNI}}, \beta, \text{mode})$; the backbone α is the only other parameter and is shared with plain regCCD.
- **No graph mutation.** The clade set is untouched so there is no clade enumeration, no second split expansion.
- **Full support.** Every tree has nonzero probability, not only those within a fixed move-set, so coverage is not capped at NNI (or any) rearrangement depth.
- **Coverage is free.** It is decoupled from the #P-hard exact normalisation: 100% coverage needs only the $O(m^3)$ low-order estimate to high accuracy for low μ .
- **Controlled normalisation.** The tail correction makes the normalisation explicit and either provably non-inflating or near-exact.

Appendix A: clade-level NNI expansion

Motivation. The CCD0 construction *expands the split set*: for every clade partition $P \rightarrow \{L, R\}$ that could exist because its parent P and both children L, R are already clades in $\mathcal{C}$ (the set of clades with nonzero marginal support), it adds that partition even if no sampled tree realised it. Here we describe the analogous operation one level up (expanding the *clade set* $\mathcal{C}$ itself) by reading nearest-neighbour-interchange (NNI) rearrangements directly off the stored splits, without ever building a tree.

NNI at the clade level. An internal edge of a tree is a stored split $P \rightarrow \{L, R\}$: the parent clade P branches into disjoint children L, R with $L \cup R = P$. An NNI at that edge needs the grandchildren. Descend into one child, say L , with its split $L \rightarrow \{A, B\}$, and let the sibling be R . The three subtrees meeting at the edge are A, B, R . The two NNI moves recombine them so that the child of P is no longer $L = A \cup B$ but

$$A \cup R \quad \text{or} \quad B \cup R. \quad (6)$$

Because $A \subseteq L$ and $L \cap R = \emptyset$, each union in (5) is a disjoint union of size ≥ 2 and a *proper* subclade of P (it omits the nonempty B resp. A). Each novel clade carries an NNI-implied *seed split*

$$(A \cup R) \rightarrow \{A, R\},$$

whose two constituents A, R are themselves clades already in $\mathcal{C}$. The roles of L and R are symmetric, so a split $R \rightarrow \{C, D\}$ likewise yields $C \cup L$ and $D \cup L$. Every quantity (A, B, R and the parent P) is read straight from the CCD’s split table.

Algorithm. For each clade P with $|P| \geq 3$, each split $\{L, R\} \in \Pi(P)$, each choice of descend-child $X \in \{L, R\}$ with sibling Y , and each grandchild split $\{A, B\} \in \Pi(X)$, emit the candidates $A \cup Y$ and $B \cup Y$; keep those not already in $\mathcal{C}$. This is a pass over pairs of adjacent splits, costing

$$O\left(\sum_P |\Pi(P)| \cdot \max_X |\Pi(X)|\right) = O(|\text{splits}| \times \bar{d}),$$

with $\bar{d}$ the average number of splits per clade, proportional to the size of the CCD, not to the number of trees it represents.

Two pairing modes. The CCD graph mixes splits from many posterior trees, so the algorithm above can combine a split $P \rightarrow \{L, R\}$ with a grandchild split $L \rightarrow \{A, B\}$ that never occurred together in any one tree. We therefore distinguish:

- **Graph-wide** ($\mathcal{N}_{\text{gw}}$): pair *any* stored split on P with *any* stored split on its child X . Cheapest and broadest; the pass described above.
- **Co-occurring** ($\mathcal{N}_{\text{co}}$): only recombine grandchild splits that actually appeared together under P in some single base tree, the true NNI neighbourhood of the posterior sample. This is obtained by walking the stored base trees and reading A, B, R off each internal edge.

By construction $\mathcal{N}_{\text{co}} \subseteq \mathcal{N}_{\text{gw}}$: the co-occurring set is the graph-wide set restricted to split pairs realised within a single tree. The graph-wide extras are exactly the clades that *no single sampled tree is one NNI move away from*; whether that is desirable broader coverage or an overreach beyond the sample’s NNI neighbourhood is a modelling choice one should state explicitly.

Conservativeness. NNI is the most local rearrangement, so the expansion recovers only clades “near” the support and misses anything requiring an SPR-scale move. For the single tree $((A, B), C), D$ the expansion yields exactly $\{ABD, CD, AC, BC\}$; it does *not* produce AD, BD, ACD or BCD , which need a non-local move. If the concern is held-out trees that differ from the sample by a single local rearrangement, this coverage is exactly right; for broader topological novelty one would extend to SPR neighbourhoods, also expressible at the clade level (a regrafted subtree clade unioned with whatever lies below the target edge), at the cost of many more candidates.

A.1 Parameterisation in CCD1 / regCCD

The clade expansion is most natural in the *split*-parameterised models. A regularised CCD (regCCD) takes the CCD1 graph, adds the unobserved splits whose parent and children are already clades (the CCD0 split expansion), and prices every split of a clade C by additive smoothing,

$$\theta_\alpha(S) = \frac{f(S) + \alpha}{f(C) + \alpha |S(C)|}, \quad (7)$$

a Dirichlet($\alpha, \dots, \alpha$) prior over the splits $S(C)$. This gives nonzero probability to trees with *unsampled splits of sampled clades*, but a tree containing an *unsampled clade* still has probability zero: the residual limitation of both CCD0 and CCD1.

NNI clade expansion closes this gap, and, crucially, needs *no new parameter*. Under the CCD1 form $P(T) = \prod_v \text{CCP}(\text{split} \mid \text{clade})$ a clade carries no marginal parameter; its probability is induced by the product along the path to it. So adding a novel clade $K = A \cup R$ only requires its *splits* to be in the graph:

- the recombined parent split $P \rightarrow \{K, B\}$ enlarges $S(P)$ and is priced by (6) with $f = 0$, i.e. mass $\alpha/(f(P) + \alpha |S(P)|)$;
- the seed split $K \rightarrow \{A, R\}$ (and any further splits of K into existing clades found by the split expansion) is priced the same way; a clade left with a single split is deterministic ($\text{CCP} = 1$).

A held-out tree that is one NNI move from the support is then scored $P(T) = [\prod \text{observed CCPs}] \cdot \theta_\alpha(P \rightarrow \{K, B\}) \cdot \theta(K \rightarrow \{A, R\}) > 0$. The novel clades enter the *same* additive- α Dirichlet-multinomial as the observed and split-expanded splits (single pseudocount), so the only model change is a larger clade set.

Coverage vs. sharpness, and a two-level prior. The benefit is out-of-sample coverage of trees with unsampled clades; the cost is that enlarging $|S(P)|$ dilutes the smoothed mass of observed splits. The latter is controlled by giving NNI-derived splits (those touching a novel clade) a *separate* pseudocount β , while observed and split-expanded splits keep α :

$$\theta(S) = \frac{f(S) + \alpha [S \text{ regular}] + \beta [S \text{ NNI}]}{f(C) + \alpha |R(C)| + \beta |N(C)|}, \quad (8)$$

an asymmetric Dirichlet placing a weaker prior on the speculative recombinations. Crucially β does not affect *coverage* (any $\beta > 0$ keeps the NNI split positive); it only rebalances mass at observed parents.

Choosing the pseudocounts. Since β does not change which trees are covered, the held-out log probability over the (fixed) covered set is a clean objective for selecting (α, β) . Because the pseudocounts do not change the graph, the model is fit once and the (α, β) grid is scored in closed form. The optimal β is typically an order of magnitude below α , confirming the weaker prior on NNI splits; and the optimal α shifts under the expansion, so the pseudocounts should be re-tuned rather than inherited.

A.2 Worked example: held-out calibration on RSV2

The most direct test of whether the expansion is well-calibrated is to train on one posterior sample and score an *independent* posterior sample from the same MCMC analysis. We use two independent runs on RSV2 ($n = 129$, each with 18,001 post-burn-in trees) and ask, at the pipeline-fitted (α, β) , how much probability mass the model places on NNI-expanded trees versus the fraction of held-out trees that actually fall in that zone.

The full pipeline. The clade set under regCCD is built in four stages,

$$\mathcal{C}^{(1)} \xrightarrow{\text{split exp.}} \mathcal{C}^{(1)} \cdot se \xrightarrow{\text{NNI exp.}} \mathcal{C}^{(1)} \cdot se \cdot ne \xrightarrow{\text{split exp.}} \mathcal{C}^{(1)} \cdot se \cdot ne \cdot se, \quad (9)$$

each enlarging the supported tree space. Crucially, the *first* split expansion runs before the NNI pass, so the NNI recombinations get to use the CCD0-expanded split contexts; this matters for graph-wide pairing, which then finds additional novel clades that the observed-only graph would miss.

Tree-space blowup. On RSV2 the four spaces have the following sizes. Both modes share $|\mathcal{C}^{(1)}| = |\mathcal{C}^{(1)} \cdot se| = 1,919$; graph-wide adds 42,627 novel clades, co-occurring adds 7,956.

stage	clades	splits	$\log_{10} N_{\text{trees}}$
$\mathcal{C}^{(1)}$	1,919	7,653	46.33
$\mathcal{C}^{(1)} \cdot se$	1,919	23,597	50.44
$\mathcal{C}^{(1)} \cdot se \cdot ne$ (gw)	44,546	157,602	71.79
$\mathcal{C}^{(1)} \cdot se \cdot ne \cdot se$ (gw)	44,546	3,142,692	87.06
$\mathcal{C}^{(1)} \cdot se \cdot ne$ (co)	9,875	41,517	67.30
$\mathcal{C}^{(1)} \cdot se \cdot ne \cdot se$ (co)	9,875	256,497	77.92

The first split expansion contributes +4.1 decades to the tree count, the NNI clade expansion contributes +21.4 (gw) or +16.9 (co), and the final split expansion contributes a further +15.3 (gw) or +10.6 (co). The NNI clade expansion is the single largest jump in tree count, while the final split expansion contributes the largest jump in raw split count (an additional $\sim 3 \times 10^6$ partitions under graph-wide), most of which are combinations of novel clades that would never appear in a sampled tree.

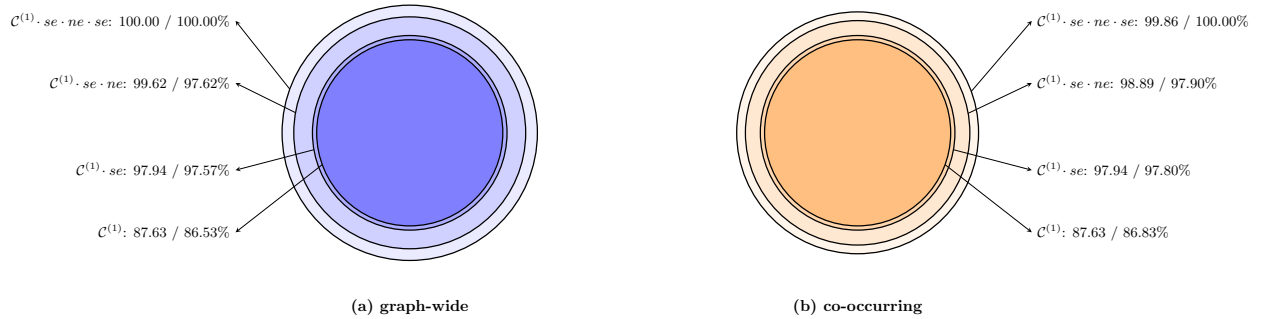


Figure 1: Tree-space envelope at each pipeline stage on RSV2 (train: run 1, 18,001 trees; held out: run 2, same size). Areas are proportional to $\log_{10} N_{\text{trees}}$ (so radii are proportional to $\sqrt{\log_{10} N}$). Each circle is annotated with the fraction of held-out trees that fall inside it (cumulative empirical coverage) and the model’s cumulative probability mass on trees inside that universe under the fitted (α, β) . The two quantities agree to within ~ 2 percentage points at every boundary in both panels, despite each transition enlarging the tree count by many decimal orders of magnitude.

Held-out fit. We train the full pipeline on run 1 and grid-search (α, β) to maximise the mean held-out log-probability on run 2 (no cross-validation needed: the two runs are genuinely independent). The optima are

mode	α^*	β^*	mean $\log P/\text{tree}$	coverage
graph-wide	0.265	0.0014	-54.21	18001/18001 (100.0%)
co-occurring	0.260	0.0084	-54.11	17976/18001 (99.86%)

The graph-wide expansion covers every held-out tree; the 25 co-occurring misses are trees whose novel clades sit more than one NNI move away from any clade-context in run 1, an SPR-scale rearrangement away.

Probability mass on NNI-expanded trees vs. empirical fraction. At the fitted (α, β) , we compute in closed form the total probability mass the model places on trees that contain at least one novel clade,

$$P(\text{novel-clade tree}) = 1 - M(\rho), \quad M(c) = \sum_{(L,R) \in S^{\text{obs}}(c)} \theta(L, R) M(L) M(R), \quad (10)$$

restricted at every clade c to the sub-sum over existing-only splits (novel children contribute $M = 0$). We then compare to the *empirical* fraction of held-out trees that actually contain a novel clade:

mode	model $P(\text{novel})$	empirical	ratio
graph-wide	2.43%	2.06%	1.18×
co-occurring	2.20%	1.92%	1.15×

The model overstates the speculative fraction by only $\sim 15\text{--}20\%$ in both modes. Given that the universe of *potentially* novel trees is $\sim 10^{40}$ times larger than the observed-clade tree set (cf. the table above), matching the empirical novel-clade rate within a factor of 1.2 is the operative sense in which the NNI clade expansion is *well-calibrated*: the additive- α, β smoothing under the pipeline of (8) carves a probability mass on the expansion zone that closely tracks the rate at which independent posterior samples actually visit it.